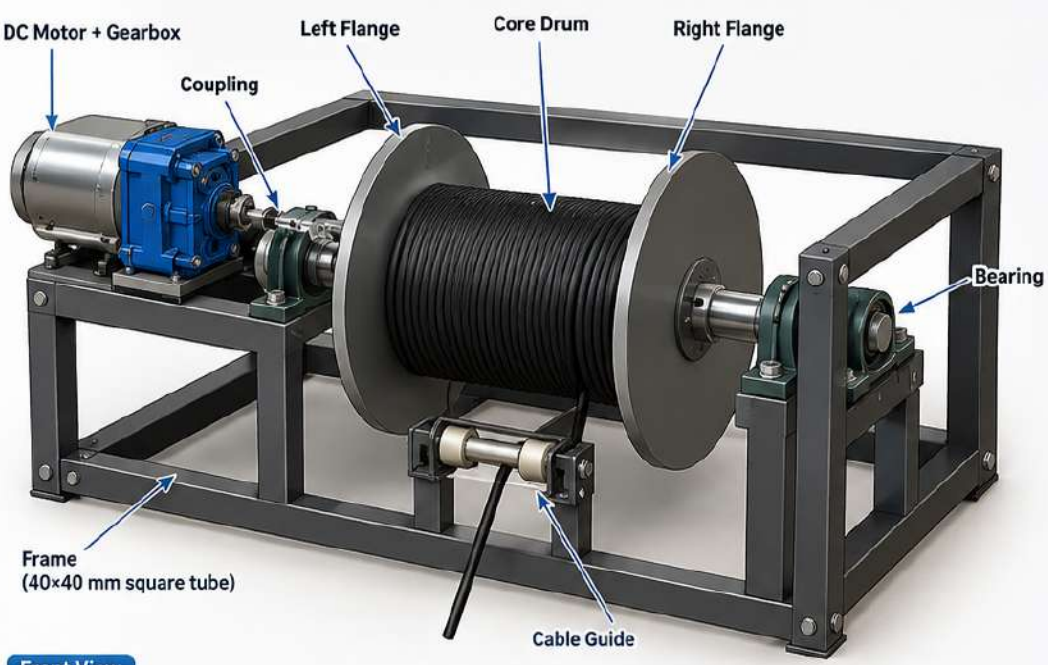


Parts List

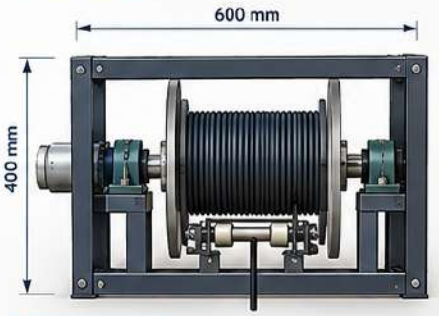
No.	Part Name	Material (Suggestion)	Qty.
1	Frame (40×40 mm square tube)	Mild Steel	1
2	Shaft (Ø20 mm)	EN8 / Stainless Steel	1
3	Core Drum (Ø200 × 300 mm)	Mild Steel / Aluminium	1
4	Flange (Ø350 mm)	Mild Steel / Aluminium	2
5	Bearing (Pillow Block UCP204)	Cast Iron	2
6	Bearing Housing	Cast Iron	2
7	Motor (12V DC)	—	1
8	Gearbox (100 kg-cm, 4 RPM)	—	1
9	Coupling (Flexible)	Steel / Aluminium	1
10	Cable Guide (Roller)	Nylon / Steel	1
11	Mounting Plate (Motor & Bearing)	Mild Steel	2
12	Bolts, Nuts, Washers	Steel	—

Key Dimensions (mm)

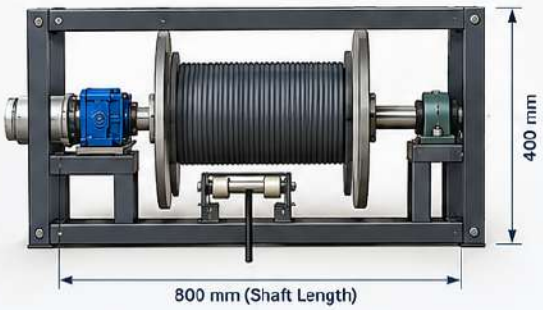
Frame Width	600
Frame Depth	400
Frame Height	400
Shaft Length	800
Core Drum Diameter (outer)	200
Core Drum Length	300
Flange Diameter	350
Shaft Diameter	20
Frame Tube Size	40 × 40

Note: Dimensions are as per your provided design. This is a conceptual assembly and may need structural verification for 32 kg cable load.

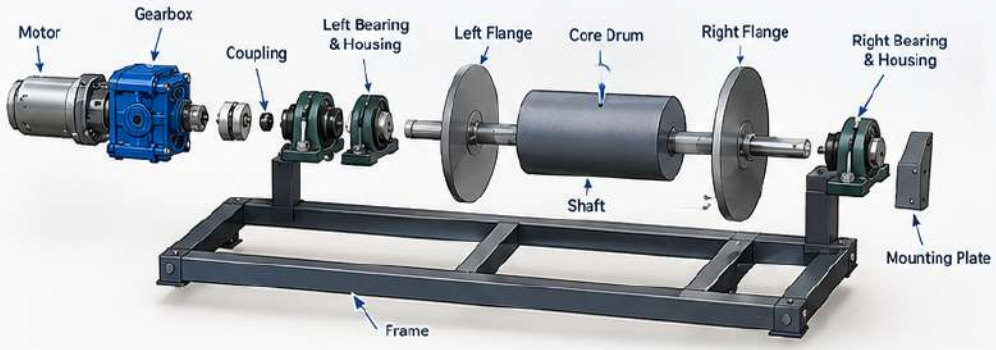
Front View



Side View



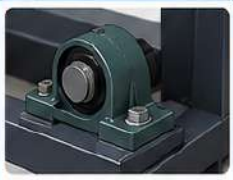
Exploded View



Key Connection Details



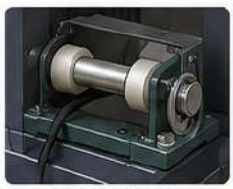
Hub / End Plate
(Drum to Shaft)



Bearing Mounting
(on Frame)

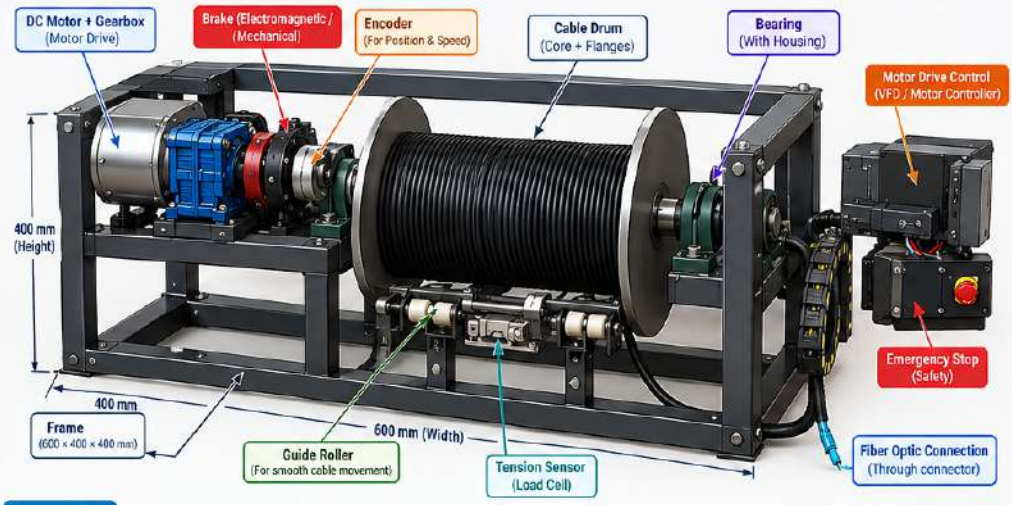


Flexible Coupling
(Motor to Shaft)



Cable Guide Roller
(Front View)

Complete Assembly – All Parts with Location



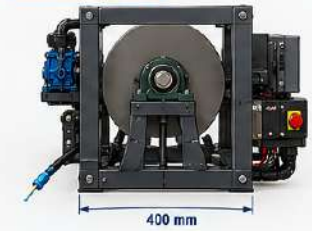
Parts List with Location

No.	Part Name	Location in Assembly	Function
1	Frame (600×400×400 mm)	Outer structure	Support for all parts
2	Shaft (800 mm)	Through drum & bearings	Rotates drum
3	Core Drum (Ø200 × 300 mm)	Center of frame	Cable winding surface
4	Flange (Ø350 mm)	Both ends of core	Prevents cable fall
5	Bearing + Housing	Left & right side of frame	Supports shaft
6	Coupling	Motor to shaft	Transfers motion
7	DC Motor + Gearbox	Left side (outside frame)	Drives drum
8	Brake (Electromagnetic)	Between gearbox & shaft	Stops drum (safe)
9	Encoder	Motor/shaft end	Position & speed feedback
10	Motor Drive Control (VFD/Controller)	On frame side	Controls motor speed/direction
11	Guide Roller	Front top/bottom	Keeps cable aligned
12	Tension Sensor (Load Cell)	Front bottom	Measures cable tension
13	Emergency Stop	Front side panel	Safety shutdown
14	Fiber Optic Connection	Cable outlet	Connects to ground station
15	Cable Guide Bracket	Front outlet	Smooth cable exit
16	Wiring / Cable Chain	Along frame	Protects cables

Front View



Side View (Right Side)



Top View



Side Dimension (Key)

Frame Width	600 mm
Frame Depth	400 mm
Frame Height	400 mm
Shaft Length	800 mm
Core Drum Diameter (outer)	200 mm
Core Drum Length	300 mm
Flange Diameter	350 mm
Shaft Diameter	20 mm
Frame Tube Size	40 × 40 mm

Note: Dimensions are as per your provided design. Final design may vary based on load and analysis.

1. Guide Roller Detail



Keeps cable aligned and prevents side rubbing.

2. Tension Sensor Detail



Measures cable tension for constant control.

3. Encoder Detail



Provides position and speed feedback to controller.

4. Brake Detail



Stops drum quickly in emergency or on stop command.

5. Motor Drive Control (VFD)



Controls motor speed, direction and torque.

6. Fiber Optic Connection



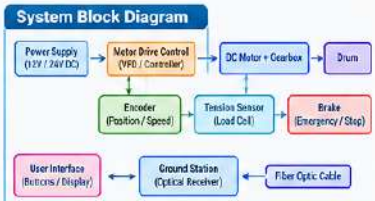
Connects to ground station (Optical data transmission).

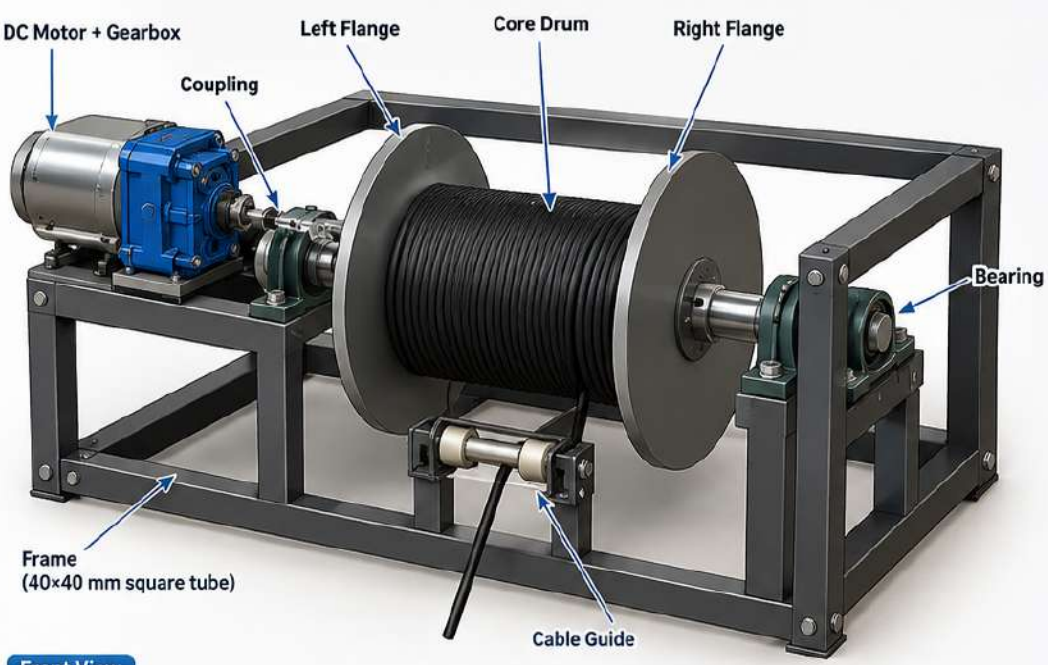
7. Emergency Stop



Instant power cut to motor drive and system.

System Block Diagram





Parts List

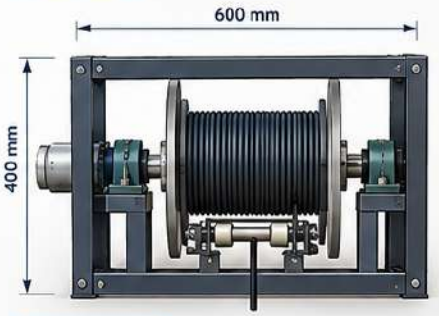
No.	Part Name	Material (Suggestion)	Qty.
1	Frame (40×40 mm square tube)	Mild Steel	1
2	Shaft (Ø20 mm)	EN8 / Stainless Steel	1
3	Core Drum (Ø200 × 300 mm)	Mild Steel / Aluminium	1
4	Flange (Ø350 mm)	Mild Steel / Aluminium	2
5	Bearing (Pillow Block UCP204)	Cast Iron	2
6	Bearing Housing	Cast Iron	2
7	Motor (12V DC)	—	1
8	Gearbox (100 kg-cm, 4 RPM)	—	1
9	Coupling (Flexible)	Steel / Aluminium	1
10	Cable Guide (Roller)	Nylon / Steel	1
11	Mounting Plate (Motor & Bearing)	Mild Steel	2
12	Bolts, Nuts, Washers	Steel	—

Key Dimensions (mm)

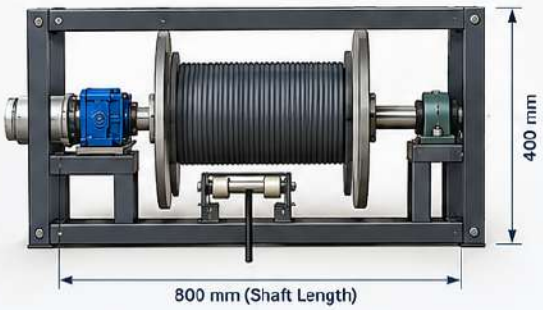
Frame Width	600
Frame Depth	400
Frame Height	400
Shaft Length	800
Core Drum Diameter (outer)	200
Core Drum Length	300
Flange Diameter	350
Shaft Diameter	20
Frame Tube Size	40 × 40

Note: Dimensions are as per your provided design. This is a conceptual assembly and may need structural verification for 32 kg cable load.

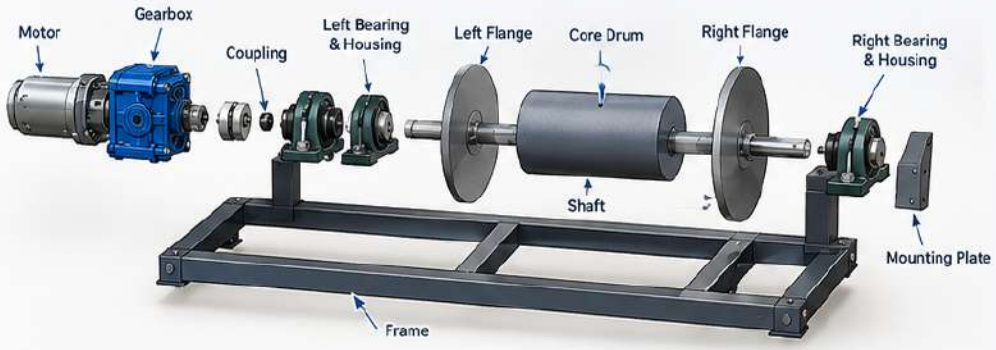
Front View



Side View



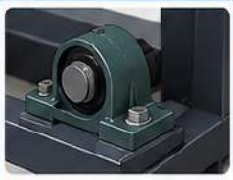
Exploded View



Key Connection Details



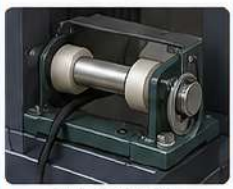
Hub / End Plate
(Drum to Shaft)



Bearing Mounting
(on Frame)



Flexible Coupling
(Motor to Shaft)

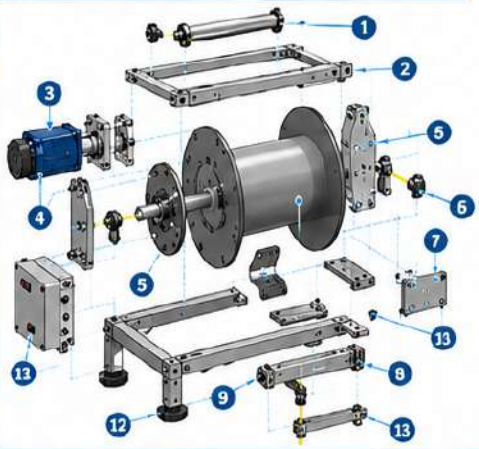


Cable Guide Roller
(Front View)

1. पूरा सिस्टम का 3D CAD मॉडल (Assembly View)



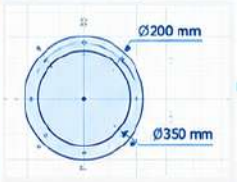
2. Exploded View (विस्फोटित दृश्य)



क्रम	पार्ट का नाम	संख्या
1	गाइड रोलर (ऊपर)	1
2	फ्रेम (टॉप सपोर्ट)	2
3	साइड प्लेट	2
4	मोटर + गियरबॉक्स	1
5	ड्रम (रोलर)	1
6	बेयरिंग + शाफ्ट	1 सेट
7	टेंशन सेंसर	1
8	कंट्रोल बॉक्स	1
9	बेस फ्रेम	1
10	सपोर्ट फुट (रबर)	4
11	माउंटिंग प्लेट	2
12	बोल्ट / नट	1 सेट
13	केबल गाइड रोलर	1

3. ड्रम का 3D मॉडल बनाना (Fusion 360)

3.1 Sketch बनाएं



- नया Sketch → Circle बनाएं
- दो सर्कल (बाहरी और आंतरिक व्यास)

3.2 Extrude करें



- बीच का सिलेंडर बनाएं (Extrude)
- लंबाई: 300 mm (उदाहरण)

3.3 Flange जोड़ें



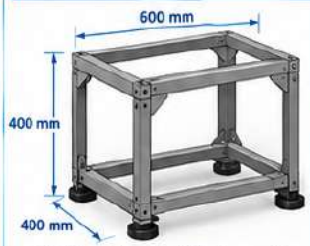
- दोनों तरफ फ्लेंज बनाएं
- मोटाई: 10 mm (उदाहरण)

3.4 Shaft Hole



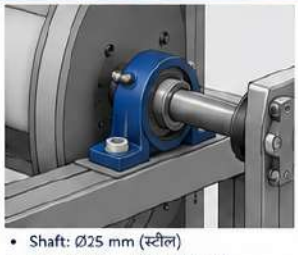
- केंद्र में Ø25 mm का छेद
- Bearing फिट करने के लिए

4. फ्रेम का 3D मॉडल



- 40×40 mm Aluminium/Steel square tube
- Corner पर gusset plate
- नीचे रबर फुट (vibration कम करने के लिए)

5. शाफ्ट और बेयरिंग असेंबली



- Shaft: Ø25 mm (स्टील)
- Pillow block bearing (UC205)
- दोनों side पर लगाएँ
- ड्रम को shaft पर फिक्स करें

6. मोटर + गियरबॉक्स माउंट



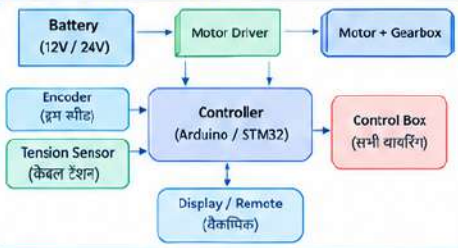
- मोटर (12V/24V DC) + Gearbox
- माउंटिंग प्लेट बनाएं
- Coupling से shaft जोड़ें

7. गाइड रोलर और टेंशन सेंसर



- केबल को सही दिशा में रखने के लिए
- टेंशन सेंसर (Load Cell) लगाएं
- रोलर की सतह smooth रखें

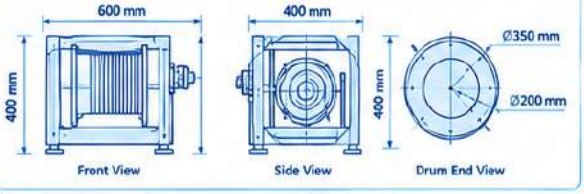
8. इलेक्ट्रॉनिक सिस्टम (सुझाव)



9. पूरा Assembly (Fusion 360)



10. महत्वपूर्ण Dimensions (उदाहरण)



11. Final Render (Realistic View)



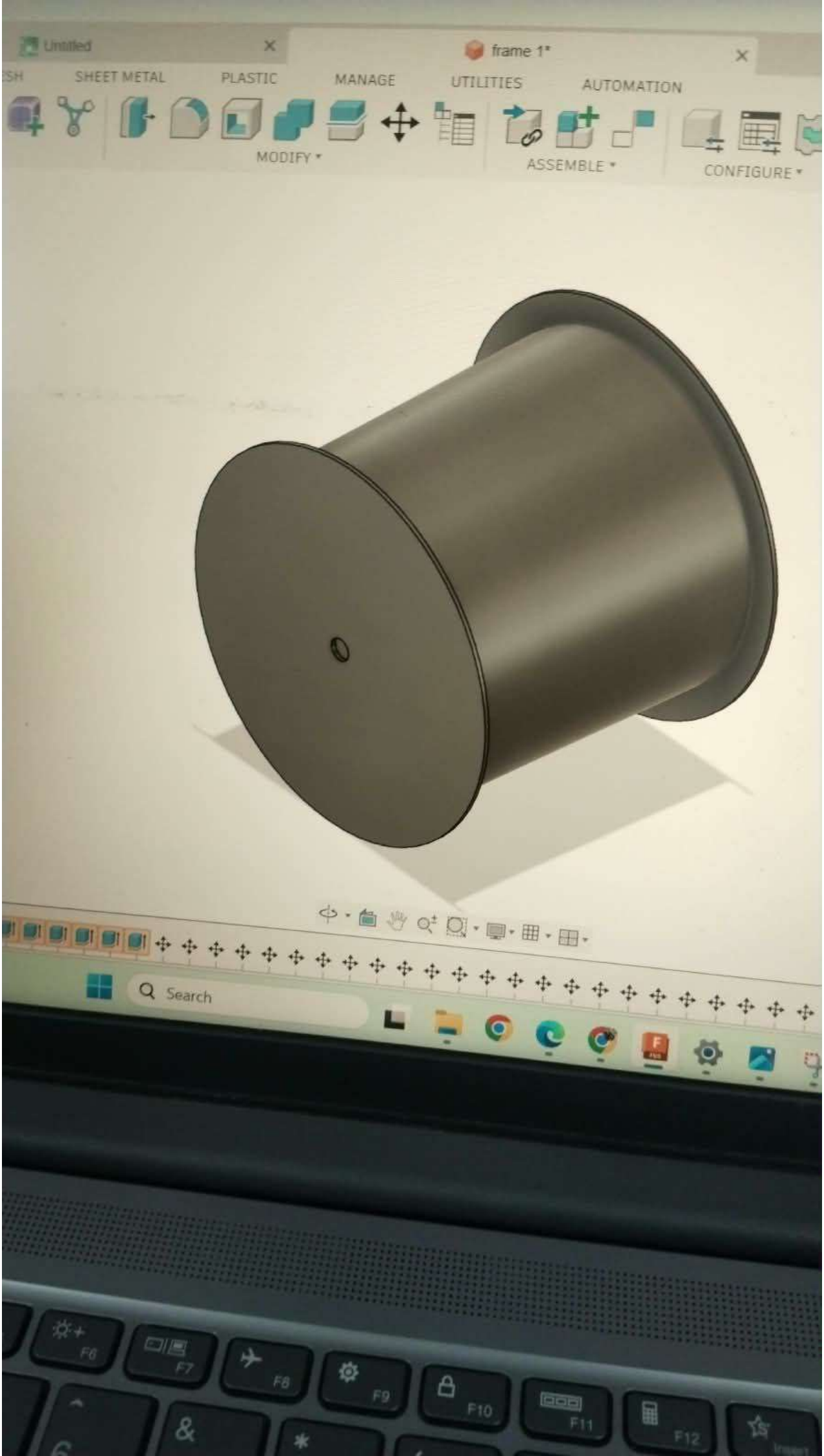
क्या ध्यान रखें?

- ✓ केबल का न्यूनतम bend radius
- ✓ ड्रम पर समान वाईंडिंग
- ✓ मोटर का टॉर्क और RPM
- ✓ टेंशन लिमिट
- ✓ वाटर/डस्ट प्रोटेक्शन (IP rating)
- ✓ सुरक्षा के लिए Emergency Stop

12. 32 kg Fiber Optic Cable के लिए सुझाव

- यदि 32 kg केबल केबल का वजन है, तो ड्रम और फ्रेम का अतिरिक्त वजन जोड़ें (कुल 50–70 kg अनुमानित).
- केबल का बाहरी व्यास, कुल लंबाई और bend radius पहले पता करें.
- ड्रोन के साथ उपयोग के लिए कुल वजन और बैलेंस बहुत महत्वपूर्ण है!
- प्रोटोटाइप पहले फिक्स्ड टेस्ट-बेंच पर बनाएं, फिर ड्रोन पर लगाएं.

यह CAD डिजाइन एक अवधारणा (Concept Design) है, जिसे अपने आवश्यकताओं और ड्रोन के प्रकार के अनुसार और बेहतर बनाया जा सकता है।


Untitled
frame 1*
SHEET METAL
PLASTIC
MANAGE
UTILITIES
AUTOMATION
MODIFY
ASSEMBLE
CONFIGURE
Search